Rody Vilchez

Applied ML Engineer — Retrieval · Agent Systems · Evaluation · Urban ML
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Applied ML engineer working at the intersection of agent systems, retrieval, and evaluation under imperfect data. At CIP–CGIAR, I build production multi-agent platforms, multilingual GraphRAG, and evaluation infrastructure; independently, I study measurement failure in urban ML and co-authored a multimodal architecture published in Springer CCIS.

Experience

Assistant — Artificial Intelligence — **International Potato Center (CIP–CGIAR)**
Lima, Peru · AI Intern, Oct 2025 – Oct 2026 — *Oct 2026 – Present*

- **Microsoft Teams multi-agent platform** — Architected and shipped an institutional system on Copilot Studio for IT support and corporate services, with cross-domain delegation and escalation to tickets prefilled from conversational context and subject to human review.
- **Agent-based evaluation suite** — Built a suite where evaluator agents execute scenarios derived from operational records; it combines deterministic regression, provider-agnostic LLM-as-judge scoring, and persona-driven simulation to attribute routing, delegation, and response-quality failures.
- **Multilingual GraphRAG** — Built end-to-end the document-intelligence layer powering a five-language agricultural GraphRAG over corpora with noisy OCR and irregular layouts: parsing, chunking, structured metadata extraction, embeddings, and vector storage.

QA Trainee — **Visma LATAM**
Lima, Peru — *Dec 2024 – Oct 2025*

- **LLM-driven test synthesis** — Built an agent that transforms functional specifications into executable E2E scenarios.
- **DOM-aware automation** — Engineered selector and state extraction, integrating generated Cypress suites into Jenkins as regression coverage for critical workflows.

Selected research and systems

Infelix — Latent Urban Crime Risk under Reporting Bias — *UPC thesis, 2025–2026 · SIMBig 2026 paper*

- **Modeling unreported crime** — Built a Bayesian model over victimization surveys and an 11-source H3 geospatial panel (Sentinel-2, VIIRS, OSM, WorldPop, and Street View) to estimate crime risk across 43 Lima–Callao districts; census data were the only source that consistently improved predictions within districts.
- **Cross-city transfer** — Features learned in Lima matched models with one year of local data in Arequipa and remained competitive in Cusco and Piura; stress tests showed that record quality, not only architecture, determines when a system can be evaluated and transferred.

Multimodal Sign Language Model — *Springer CCIS 2895, 2026*

- Co-authored a published gloss-free multimodal architecture using latent queries and cross-attention to project 2D keypoints into a pretrained language model's embedding space, evaluated across Peruvian and Argentinian corpora. DOI: 10.1007/978-3-032-20322-9_23

Wachi — Risk-Aware Pedestrian Routing — *El Comercio Lab · wachi.rosewt.dev*

- Built a pedestrian router for Lima that pairs the shortest route with a calmer one past fewer registered robberies and thefts within 150 m, offered only if it cuts that exposure by at least 20% within a detour of +15% or +5 min.

Open-source contributions

- **Microsoft Copilot Studio** — Diagnosed knowledge-source synchronization failures in child agents and validated prerelease fixes in the official VS Code extension.
- **Gemini CLI** — Corrected a UI schema and added regression coverage to Google's open-source agentic CLI.
- **scikit-learn** — Improved HDBSCAN technical documentation.
- **beads (bd)** — Implemented prefix-routed write-through and the regression test that unlocked consolidation of the upstream implementation.

Education and recognition

B.Sc. Computer Science — **Universidad Peruana de Ciencias Aplicadas (UPC)**
Expected late 2026
2nd place, DataFest BCP × ESAN 2025 · SALA 2026 full scholarship

Technical focus

Retrieval and agents: LlamaIndex, Qdrant, Copilot Studio, Power Automate, multi-agent orchestration
Evaluation and research: Python, PyTorch, scikit-learn, LLM-as-judge, experimental design, robustness and uncertainty
Data and backend: FastAPI, H3, DuckDB, PostgreSQL, ETL, REST APIs, Docker, GitHub Actions, Linux
Languages: Spanish (native) · English (intermediate)